# AHP Decision Analysis Report

## Laptop Selection

**Goal:** Select the optimal laptop considering performance, price, and portability

**Generated:** 9/26/2025

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## Executive Summary

**Best Alternative:** MacBook Pro 16" (Score: 0.3356)

**Number of Criteria:** 3

**Number of Alternatives:** 4

**Overall Consistency:** **Acceptable (CR = 0.0081)**

## Final Rankings

| Rank | Alternative | Score | Percentage |
|---|---|---|---|
| 1 | MacBook Pro 16" | 0.3356 | 33.56% |
| 2 | ThinkPad X1 Carbon | 0.2732 | 27.32% |
| 3 | Dell XPS 15 | 0.2471 | 24.71% |
| 4 | Surface Laptop Studio | 0.1441 | 14.41% |

## Criteria Weights

| Criterion | Weight | Percentage | Consistency Ratio |
|---|---|---|---|
| Performance | 0.6483 | 64.83% | 0.0054 |
| Price | 0.1220 | 12.20% | 0.0249 |
| Portability | 0.2297 | 22.97% | 0.0283 |

## Alternative Scores by Criteria

### Performance

| Alternative | Score | Percentage |
|---|---|---|
| MacBook Pro 16" | 0.4832 | 48.32% |
| Dell XPS 15 | 0.2717 | 27.17% |
| ThinkPad X1 Carbon | 0.0882 | 8.82% |
| Surface Laptop Studio | 0.1569 | 15.69% |

### Price

| Alternative | Score | Percentage |
|---|---|---|
| MacBook Pro 16" | 0.0571 | 5.71% |
| Dell XPS 15 | 0.2469 | 24.69% |
| ThinkPad X1 Carbon | 0.5531 | 55.31% |
| Surface Laptop Studio | 0.1430 | 14.30% |

### Portability

| Alternative | Score | Percentage |
|---|---|---|
| MacBook Pro 16" | 0.0667 | 6.67% |
| Dell XPS 15 | 0.1778 | 17.78% |
| ThinkPad X1 Carbon | 0.6468 | 64.68% |
| Surface Laptop Studio | 0.1086 | 10.86% |

## Consistency Analysis

The Analytic Hierarchy Process includes consistency checking to ensure the reliability of judgments.

- **Consistency Ratio (CR) ≤ 0.1:** Acceptable consistency
- **Consistency Ratio (CR) > 0.1:** Inconsistent judgments, review recommended

**Overall Consistency Ratio:** 0.0081

## Methodology

This analysis used the Analytic Hierarchy Process (AHP), a multi-criteria decision analysis method developed by Thomas Saaty. The process involves:

1. **Hierarchy Construction:** Decomposing the decision into goal, criteria, and alternatives
2. **Pairwise Comparisons:** Comparing elements using Saaty's 1-9 scale
3. **Priority Derivation:** Calculating weights using the eigenvector method
4. **Consistency Verification:** Ensuring logical consistency of judgments
5. **Synthesis:** Combining weights to determine final priorities

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